

Foundation (Jalapeno)

Q1. Factor the trinomial $x^2 + 5x + 6$

- (A) $(x + 2)(x + 3)$
- (B) $(x + 1)(x + 6)$
- (C) $(x + 3)(x + 2)$
- (D) $(x - 2)(x - 3)$
- (E) $(x - 1)(x - 6)$

Q2. A space lab experiment requires a trinomial equation $x^2 - 7x + 12$ to model planet orbits. Factor it:

- (A) $(x - 3)(x - 4)$
- (B) $(x + 3)(x + 4)$
- (C) $(x - 2)(x - 6)$
- (D) $(x + 2)(x + 6)$
- (E) $(x - 1)(x - 12)$

Q3. The gravity on a planet is modeled by $2x^2 + 5x + 2$. Factor the trinomial:

- (A) $(2x + 1)(x + 2)$
- (B) $(2x - 1)(x - 2)$
- (C) $(2x + 2)(x + 1)$
- (D) $(x + 1)(2x + 2)$
- (E) $(x + 2)(2x + 1)$

Q4. Factor the trinomial $x^2 - 9x + 20$

- (A) $(x - 4)(x - 5)$
- (B) $(x + 4)(x + 5)$
- (C) $(x - 2)(x - 10)$
- (D) $(x + 2)(x + 10)$
- (E) $(x - 1)(x - 20)$

Q5. In a physics experiment, the trajectory of a rocket is modeled by $x^2 + 2x - 15$. Factor:

- (A) $(x + 5)(x - 3)$
- (B) $(x - 5)(x + 3)$
- (C) $(x + 2)(x - 8)$
- (D) $(x - 2)(x + 8)$
- (E) $(x - 1)(x + 15)$

Q6. A planet's atmospheric pressure is modeled by $3x^2 - 2x - 5$. Factor the trinomial:

- (A) $(3x + 1)(x - 5)$
- (B) $(3x - 1)(x + 5)$
- (C) $(3x - 5)(x + 1)$
- (D) $(3x + 5)(x - 1)$
- (E) $(x + 1)(3x - 5)$

Q7. The temperature on Mars is modeled by $2x^2 - 5x - 3$. Factor the trinomial:

- (A) $(2x + 1)(x - 3)$
- (B) $(2x - 1)(x + 3)$
- (C) $(x + 1)(2x - 3)$
- (D) $(x - 1)(2x + 3)$
- (E) $(2x - 3)(x + 1)$

Q8. The growth of a black hole is modeled by $x^2 + 4x + 4$. Factor the trinomial:

- (A) $(x + 2)^2$
- (B) $(x - 2)^2$
- (C) $(x + 1)^2$
- (D) $(x - 1)^2$
- (E) $(x + 2)(x - 2)$

Open Ended Questions (FRQ)

Q9. Factor the trinomial $x^2 - 6x - 16$ and explain your steps.

Q10. The distance between two planets is modeled by $2x^2 + 7x + 3$. Factor the trinomial and describe a real-world scenario where this equation would be used.

Chef's Tips

- Always check for common factors before factoring.
- Use the FOIL method to check your factoring.
- Encourage students to draw diagrams to visualize the factoring process.